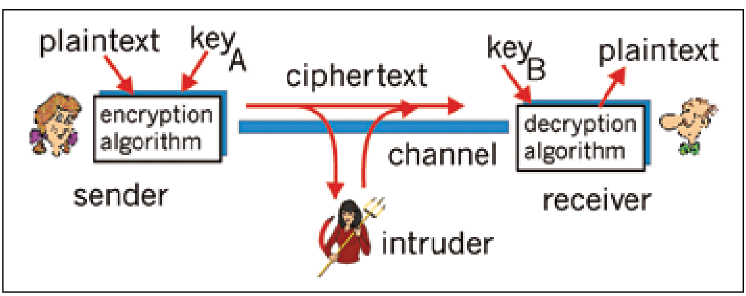
with ciphers were used throughout history right up to modern times. These devices underwent further changes when computers and electronics came into the picture, immensely helping in cryptanalysis. Cryptography has become more mathematical now and also finds applications in day-to-day security. It helps you safely transfer or withdraw money electronically and you'd be hard-pressed to come across an individual without a credit or debit card. The public-key encryption system introduced the concept of digital signatures and electronic credentials. Cryptography has a definitive existence in our lives today and the whole system will crumble in its absence.

Let's now discuss the varied uses of cryptography in modern times and its intersection with computer science.

Secrecy in transmission

The major goal of cryptography is to prevent data from being read by any third party. Most transmission systems use a private-key cryptosystem. This system uses a secret key to encrypt and decrypt data which is shared between the sender and receiver. The private keys are distributed and destroyed periodically. One must secure the key from unauthorized access, because any party that has the key can decrypt the encrypted information. Alternately a key-generating-key, called a master key, can be used to electronically generate a one-time session-key for every transaction. The secrecy of the master-key should be maintained by all parties privy to the information. The disadvantage of this method is there's too much hope riding on the master-key, which if cracked, collapses the entire system.

A better method is to use a public-key cryptosystem. In this system, data can be encrypted by anyone with the public-key, but it can be decrypted only by using the private-key, and data that is signed with the private key



How an encrypted transmission can be intercepted